Probabilistic Game Theory and Stochastic Model Predictive Control-Based Decision Making and Motion Planning in Uncontrolled Intersections for Autonomous Driving

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***Abstract*—This paper presents a motion planner of autonomous vehicles for uncontrolled intersection driving. Uncertainty about the surrounding target vehicles is an important consideration for decision-making and motion planning for autonomous driving. An interactive Multiple Model (IMM) filter is used to estimate the maneuver probability of target vehicles by using the measurement of environment sensors. An intelligent driver model is utilized to reflect the asymmetry of the longitudinal acceleration to a local model of the IMM filter. Estimates of the drivers' intentions are used to estimate the probability of the possible driving scenes. A game theory-based supervisor is configured based on the probability of the driving scene and uncontrolled intersection theory to determine the driving mode. A payoff for each driving scene is designed to consider the risk and traffic efficiency based on the prediction of the targets. Local filters of IMM are used to predict the future motion of the targets for evaluation of the payoff function. A stochastic model predictive controller is utilized to determine the desired longitudinal acceleration. Chance constraints are defined to reflect the uncertainty of the target prediction in motion planning. The proposed algorithm is evaluated via case studies and Monte Carlo simulation. The simulation results show the probabilistic game theory-based motion planner achieves the safety and efficiency of autonomous intersection passing.**

***Index Terms*—Autonomous vehicle, decision making, game theory, stochastic model predictive control (SMPC), motion planning.**

I. Introduction

THE introduction of the active safety system improves vehicle safety by providing warning and assistance for driving. The success of active safety has led to research on functions for driver's convenience [1]. Lane-Keeping Assistant System (LKAS) and Smart Cruise Control (SCC) are the representative functions for lateral and longitudinal assistance, respectively. These functions are called Advanced Driver Assistant Systems (ADAS). In recent years, ADAS, which is level 2 automation, has been widely adopted on a variety of vehicle models [2], [3]. The experience of driving with ADAS invokes a higher level of driving automation, leading to the introduction of level 3 autonomous driving in the market. Additionally, research has been extended to fully autonomous driving to improve convenience and achieve zero accidents [4]. Initially, research on autonomous driving began on motorways and gradually expanded to urban and unconstructed environments [5]. Research on autonomous parking assistant systems also began around the same time. Furthermore, the concept of door-to-door autonomous driving system, from highways to the parking lot, was proposed by integrating all functions of ADAS.

Various methodologies have been developed to realize autonomous driving in urban environments [6], [7]. The key issue of urban autonomous driving is recognizing the complicated driving scene with various types of traffic participants. In particular, the interaction of vehicles should be considered to provide the appropriate behavior of the Autonomous Vehicle (AV). Decision-making for uncontrolled intersections is one of the important issues for autonomous driving on urban roads. Since there is no traffic signal to guide the vehicles, the desired motion of the AV should be determined based on the driving scene awareness.

A rule-based decision-making algorithm was proposed to model intersection crossing rules. First in first out is the basic rules for uncontrolled intersections. After detecting the target vehicle, the worst conflict point is searched based on the intersection structure and the desired route of the AV. Then, the time to merge is calculated to determine the crossing priority of the AV [8]. Similarly, vehicle-to-vehicle communication was used to acquire the target information for priority decisions based on traffic rules [9], [10]. Formal logic sentences were utilized to acquire flexibility by separating the mathematical model from traffic rules [11]. The ontology-based knowledge base was proposed to consider not only the traffic regulations but also maps for decision-making [12]. The state machine was designed to configure the driving mode decision module [13]. However, traffic rule-based algorithms have the disadvantage

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of being vulnerable to vehicles that violate traffic laws. To address this issue, fuzzy control was utilized in a study to model uncertain target vehicle states [14]. In another study, the priority of the vehicle was determined using the first-come-first-serve rule, and then Model Predictive Control (MPC) was used to optimize vehicle motion considering longitudinal-lateral dynamics [15]. In a study that assumed connected vehicles, game theory was used for decision-making at an uncontrolled roundabout, which considered personalized driving style, and MPC was used to obtain the optimal motion [16]. Nash bargaining game was used instead of the Stackelberg game, with a similar approach using MPC [17]. However, a rule-based approach may not make appropriate decisions when vehicles violate traffic rules or exceptions occur. In other words, the complexity and uncertainty of the traffic environment are difficult to respond to with a rule-based approach, especially when the intersection structure is complicated, and the number of vehicles to be considered increases. In this case, it is difficult to design appropriate rules.

To compensate for the uncertainty of the target vehicles, a probabilistic approach was introduced to infer their driving intention. Hidden Markov Model (HMM) was utilized to model driving behavior and provide prior information for rule-based decision-making [18]. However, the perception algorithm could not estimate all states of surrounding targets. To account for the limitations of target state estimation, the Partially Observable Markov Decision Process (POMDP) was utilized to decide the desired driving mode [19], [20], [21]. In particular, Long Short-Term Memory (LSTM) cell was used to enhance the modeling accuracy of the temporal dependency of vehicle behavior [20]. The HMM and POMDP were integrated to design the intention prediction-based decision-making [22]. Bayesian networks were utilized to estimate the possibility of collision with surrounding targets. Information on traffic signals, time to enter, Time to Collision (*TTC*), and estimated risk were considered to decide the AV's maneuver [23]. From the perspective of traffic engineering, methodologies were proposed for optimizing the entire vehicle [24] or controlling the platoon [25] under the assumption that all vehicles are connected. Many probabilistic approaches have been based on the HMM and POMDP. However, the HMM requires every state and its corresponding observed object, which is difficult to obtain using autonomous vehicle sensors while driving. The POMDP solves the problem of unobservable information but has the disadvantage of being computationally intractable to finding the optimal policy.

A Support Vector Machine (SVM) was combined with the Gaussian Mixture Model (GMM) and HMM to improve the accuracy of intention prediction. Recurrent Neural Networks (RNN) with Variational Auto Encoder (VAE) were used to learn the temporal dependency of the vehicles [26]. Since the intersection driving situation constitutes a small portion of the entire driving situation, collecting driving data is challenging. To address the data collection problem, reinforcement learning was introduced for intersection motion planning. A Deep Q-Network (DQN) was utilized to decide the driving mode at intersections. [27], [28]. Reinforcement learning using the latent inference by supervised learning was designed to consider the additional prior knowledge at training [29].

The interaction of the human-driven vehicle can be considered as a game with agents, which only have limited information. Game theory-based motion planning has been applied to find the optimal driving mode for lane change [30], merging, yielding at ramps [31], and car following [32]. Game theory has also been used to deal with negotiations between vehicles entering intersections. A level-k game theory was utilized to model vehicle behavior in unsignalized intersections as a hierarchy of levels of iterated rationalizability [33]. A leader-follower relationship was used to design the game considering driving rules, perception range, and deadlock [34]. Nash equilibria-based decision-making was proposed for AVs in unsignalized intersections [35], [36]. The integrated approach of level-k reasoning and deep reinforcement learning was used to obtain the adaptive policy for the decision-making module [37]. To deal with the limited perception range of the autonomous vehicle, wireless communication has been introduced to provide information about the targets in occluded regions [38], [39], [40].

Various methodologies have been used to determine the driving mode and desired motion for autonomous driving in uncontrolled intersections. The issues with autonomous driving in uncontrolled intersections can be summarized in four ways. First, it is difficult to create rules that encompass all situations that arise at uncontrolled intersections. Since not all drivers follow driving rules, rule-based decision-making algorithms are susceptible to actual driving situations. Additionally, when multiple vehicles enter an intersection, it is challenging to establish rules to deal with complex driving conditions. Even if intention inference and prediction are introduced, the complexity of the prediction increases when considering interactions between multiple targets approaching the intersection. The second issue is that there is limited time to make decisions and control inputs. Intersections have a smaller detection range than roads and intersecting driving routes, so there is insufficient time to accumulate time-series data. Consequently, many probabilistic approaches, such as POMDP, and sequence-based approaches, such as RNN, are not practical in real-time.

The third challenge is constructing the training dataset. Intersections account for a small portion of the road network. Most learning-based approaches require a large number of datasets. Intersections present numerous scenarios that may arise, yet they only comprise a small percentage of all roads. This makes it challenging to obtain a diverse dataset that is essential for learning. The last challenge is the difficulty of setting the input structure due to the variation in the number of vehicles to be considered. To take into account the driving situation, it is necessary to consider the interaction of various numbers of vehicles. It is difficult to design a learning-based model to handle various numbers of targets. That is, it is challenging to design an artificial neural network for situations where the number of input targets varies.

In this paper, a probabilistic game theory and Stochastic Model Predictive Control (SMPC)-based motion planner in uncontrolled intersections for AV is proposed. Game theory is used to design the supervisor of the algorithm and model the decision-making of the human drivers. To consider target uncertainty, a prediction for targets is introduced to enhance

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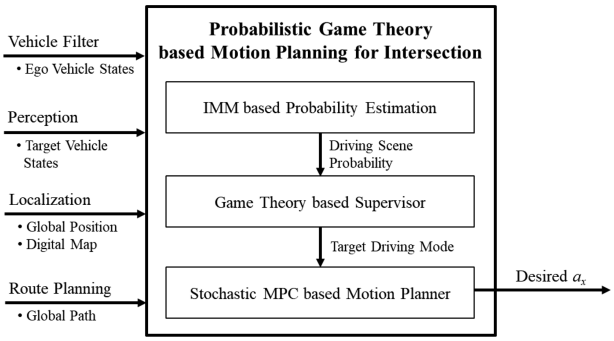

Fig. 1. Overall architecture of the probabilistic game theory-based intersection motion planning algorithm.

the accuracy of the driving mode decision of the supervisor. A new target classification is proposed to prevent an increase in prediction complexity due to multi-vehicle consideration. Additionally, Interactive Multiple Model (IMM) filter based on an intersection map is proposed to reduce the computational burden and improve the responsibility for probability estimation. The prediction results of each target are used to estimate the probability of possible driving scenes. Although the prediction is used for the supervisor, there is a possibility that risk situations due to the prediction uncertainty may occur. Therefore, it is possible to pass intersections more safely and efficiently when the state-level prediction results are reflected in the motion planner. This study introduces an SMPC to consider the prediction uncertainty of the targets [41]. The simulation study demonstrated the effectiveness of the proposed algorithm.

The main contributions of the paper are three-fold:

1) A new target classification-based game is designed to consider complex multi-target scenarios and respond to varying numbers of target vehicles.
2) The supervisor is based on driving scene probability and state-level prediction of targets using IMM to reflect the driver's decision-making process and reduce estimation time.
3) SMPC is introduced to explicitly consider uncertainty in target state prediction and improve the robustness of motion planning.

The remains of this paper are organized as follows: An overall architecture of the proposed algorithm is presented in Section II. An estimation for driving scene probability is described in Section III. Sections IV and V present the supervisor for driving mode decision based on game theory and SMPC-based motion planner. The simulation results are described in Section VI. The conclusion and the future works of this study are summarized in the final section.

II. Overall Architecture

This study focused on the prediction and motion planning of AV for safe driving in uncontrolled intersections. The overall architecture of the proposed motion planner is described in Fig. 1. This study uses four upper modules to acquire information for the motion planner. The vehicle filter, perception, and localization module are utilized to process sensor measurements. Route planner provides the global path to guide the AV in intersections. Based on the outputs from the upper modules, the motion planner decides the desired longitudinal acceleration. To secure safety and traffic efficiency, the proposed algorithm consists of three sub-modules: (1) IMM-based probability estimator, (2) game theory-based supervisor, and (3) SMPC-based motion planner. The first module is designed based on an IMM filter to estimate the probability of the driving scene. This study considers the all-possible driving scene, which is defined by the combination of the possible maneuvers of the target vehicles. To search for the optimal driving mode, game theory is utilized to convert the interaction of the vehicles into a game. The probability of the driving scene is reflected in the payoff function of the game to consider the probabilistic behavior of the vehicles with safety indices. After the driving mode is determined, the final module determines the control input using the SMPC approach. The uncertainty of the prediction is considered a chance constraint of SMPC.

III. Driving Scene Probability Estimation

The driving scene at uncontrolled intersections is composed of the various target vehicles, that interact with each other. However, modeling all interactions to predict the future driving scene leads to an exponential increase in computational burden. Thus, this study focuses on the behavior of the individual vehicle, as the current motion of a vehicle is the result of the driver's interpretation of the driving scene. Moreover, human drivers are prone to misjudgments. such as careless monitoring of traffic or misinterpreting the intentions of other drivers. Additionally, the interaction model in the driving scene can only be applied when the Field of View (FOV) of all vehicles is available. However, due to blind spots in the intersection environment, it is difficult to recognize all vehicles involved in the current traffic situation using sensors mounted on AVs. Therefore, this study estimates the maneuver probability of each vehicle separately, and the driving scene is defined by the combination of the vehicle's maneuvers. Consequently, the probability of the driving scene is determined by multiplying the probability of each target vehicle.

A. Intersection Behavior Model

The first step of the probability estimation for the vehicle's behavior is defining the maneuver sets. The driving route through intersection depends on the structure of the road, which is represented in the intersection map shown in Fig. 2. The intersection map includes the position of the input and output nodes, marked as blue and red circles, respectively, as well as the intersection zone and lanes. Among all paths connecting the input and output nodes, the driving path considered in this study is shown in Fig. 2.

Based on the intersection map, the maneuver is defined by the combination of the driving routes and longitudinal motion. The driving routes consist of three candidates, turning left, turning right, and going straight. If the structure of the intersection is changed, a different number of lateral motions can be defined. Unlike driving on main roads, the vehicle, which tries to pass the intersection, shows a variety of behaviors from a complete stop

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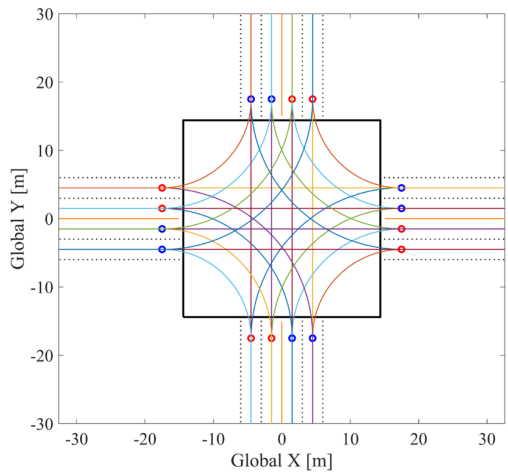

Fig. 2. Intersection map with driving routes.

to passing without braking. To model the wide range of longitudinal motion, many models are required. However, the higher the number of models, the greater the increase in computation. Thus, this study only uses two models, the cross and stop model to simplify the longitudinal motion. Therefore, the number of maneuvers is six for each vehicle.

The maneuver model is defined based on the point mass model with lateral and longitudinal controllers. It is assumed that the body slip angle is maintained at a negligible level, and thus the vehicle motion can be represented as follows:

$$\begin{aligned} x(k+1) &= x(k) + \cos(\theta(k)) \cdot v(k) \cdot dt \\ y(k+1) &= y(k) + \sin(\theta(k)) \cdot v(k) \cdot dt \\ \theta(k+1) &= \theta(k) + \gamma(k) \cdot dt \\ v(k+1) &= v(k) + a(k) \cdot dt \end{aligned} \tag{1}$$

where $x(k)$, $y(k)$, and $\theta(k)$ are the x position, y position, and heading angle with respect to the intersection coordinate system. The origin of the intersection coordinate system is located at the center of the intersection. The x-axis of the intersection coordinate system is aligned with one of the branches. $v(k)$ is the longitudinal velocity of the vehicle. dt is the sampling time of the probability estimator.

A yaw rate $\gamma(k)$ and acceleration $a(k)$ are the input from the controller to follow the driving route given in the intersection map and control the velocity of the vehicle. To determine the $\gamma(k)$, the preview point with a preview time of 1.2 s and the position of the vehicle is used to define the 2nd order polynomial $y = g(x)$, which connects the preview point on the driving route and current vehicle position. The heading constraint is applied at the current vehicle position. Since the slip of the vehicle is ignored, the vehicle heading and the 2nd-order polynomial are in contact. The radius of curvature $r(k)$ and $\gamma(k)$ is calculated as follows:

$$\begin{aligned} r(k) &= \left| \frac{(1+\dot{g}^2)^{3/2}}{\ddot{g}} \right| \\ \gamma(k) &= \frac{v(k)}{r(k)} \end{aligned} \tag{2}$$

In determining $a(k)$, the asymmetry of longitudinal acceleration should be considered. In general, the driver uses a larger deceleration than acceleration. An Intelligent Driver Model (IDM) was used to reflect these characteristics in motion prediction [42]. By using the IDM, $a(k)$ is calculated as follows:

$$\begin{aligned} a(k) &= a_{\max}\left(1 - \left(\frac{v(k)}{v_{des}}\right)^{\delta} - \left(\frac{s^*(v(k), \Delta v(k))}{s}\right)^2\right) \\ where\ s^*(v(k), \Delta v(k)) &= s_0 + v(k) \cdot T + \frac{v(k) \cdot \Delta v(k)}{2\sqrt{a_{\max} b}} \end{aligned} \tag{3}$$

where, $a_{\max}$, b, and δ are the maximum acceleration, comfortable braking deceleration, and acceleration exponent. $s^*(v,\Delta v)$ and v_{des} are the clearance with the desired headway time T and the desired velocity of AV. Δv is the relative velocity between the ego and the preceding vehicle. s and s_0 are the current and minimum clearance with a preceding vehicle.

The IDM has two components: one for velocity tracking and another for preceding vehicle following. Different desired values are used for each component based on the driving mode. In cross mode, the speed limit of the road is used as v_{des}, and the clearance is calculated based on information about the preceding vehicle. In stop mode, a virtual preceding vehicle is generated at the input node with zero velocity. The desired acceleration for stop mode is calculated using (3) to safely stop behind the virtual preceding vehicle. Since the IMM based on the behavior model should cover most of the behavior that occurs at the intersections, v_{des} in cross and stop mode are set to have a large difference.

B. IMM-Based Probability Estimation

The probability of the maneuvers is estimated by using the IMM filter. The local filter of the IMM is designed based on the intersection behavior model. The number of local filters is determined by searching for possible drivable routes from the intersection map. Thus, the IMM filter uses an intersection map and vehicle states from the environment sensors. The first step is defining the state transition and observation models for the local Kalman filter. The state vector $\mathbf{x}(k)$ and input vector $\mathbf{u}(k)$ are defined as follows:

$$\begin{aligned} \mathbf{x}(k) &= \begin{bmatrix} x(k) & y(k) & \theta(k) & v(k) \end{bmatrix}^T \\ \mathbf{u}(k) &= \begin{bmatrix} \gamma(k) & a(k) \end{bmatrix}^T \end{aligned} \tag{4}$$

By substituting (2) and (3) in (1), the nonlinear state transition model f is defined. As given in (4), the state vector of the IMM filter consists of measurements from the environment sensors. Thus, the observation model h is defined as an identity function.

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{x}(k) + \begin{bmatrix} \cos(\theta(k)) \cdot v(k) \cdot dt \\ \sin(\theta(k)) \cdot v(k) \cdot dt \\ \gamma(k) \cdot dt \\ a(k) \cdot dt \end{bmatrix} + \mathbf{w}(k) \\ &= f(\mathbf{x}(k), \mathbf{u}(k)) + \mathbf{w}(k) \\ \mathbf{z}(k) &= \mathbf{x}(k) + \mathbf{v}(k) \\ &= h(\mathbf{x}(k)) + \mathbf{v}(k) \end{aligned} \tag{5}$$

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where $\mathbf{w}(k)$ and $\mathbf{v}(k)$ are the process and measurement noises, respectively. Since the state transition model is a nonlinear function, the prediction of the local Kalman filter is defined as an extended Kalman filter. Therefore, the prediction of the local Kalman filter is derived as follows:

$$\begin{aligned}\hat{\mathbf{x}}(k+1|k) &= f\left(\hat{\mathbf{x}}(k|k), \mathbf{u}(\mathbf{k}|\mathbf{k})\right)\\ \mathbf{P}(k+1|k) &= \mathbf{F}(k)\mathbf{P}(k|k)\mathbf{F}(k) + \mathbf{Q}(k)\\ where\ \mathbf{F}(k) &= \frac{\partial f}{\partial \mathbf{x}}|_{\hat{x}(k|k),u(k)}\end{aligned} \tag{6}$$

where $\mathbf{P}$ and $\mathbf{Q}$ are the estimated covariance matrix of the state vector and the covariance of the process noise. The update of the local Kalman filter is derived as follows:

$$\begin{aligned}\widetilde{\mathbf{y}}(k) &= \mathbf{z}(k) - \hat{\mathbf{x}}(k+1|k)\\ \mathbf{S}(k) &= \mathbf{H}(k)\mathbf{P}(k+1|k)\mathbf{H}(k)^T + \mathbf{R}(k)\\ \mathbf{K}(k) &= \mathbf{P}(k+1|k)\mathbf{H}(k)^T\mathbf{S}(k)^{-1}\\ \hat{\mathbf{x}}(k+1|k+1) &= \hat{\mathbf{x}}(k+1|k) + \mathbf{K}(k)\widetilde{\mathbf{y}}(k)\\ \mathbf{P}(k+1|k+1) &= \left(I - \mathbf{K}(k)\mathbf{H}(k)\right)\mathbf{P}(k+1|k)\\ where\ \mathbf{H}(k) &= \frac{\partial h}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}(k+1|k)}\end{aligned} \tag{7}$$

where $\mathbf{R}$(k) is the covariance of the measurement noise. As mentioned in Section III-A, the six maneuvers are considered for the target vehicle. Thus, the number of the local filter is six. To infer the intention of the target vehicle, the probability of the local filter is estimated by the interaction of the IMM filter. Through the interaction, filtering, and combination steps of the IMM filter, the model probability of j-th model at time k is evaluated. Since the filtering is almost the same as in (6) and (7), the additional process of the IMM filter is defined as follows:

$$\begin{aligned}&Interaction:\\ &\bar{c}_j = \sum_i p_{ij}\mu_i(k-1)\\ &\hat{\mathbf{x}}_{0j}(k|k) = \sum_i \hat{\mathbf{x}}_i(k|k)\mu_{i|j}(k|k)\\ &\mathbf{P}_{0j}(k|k) = \sum_i \Big\{P_i(k|k)\\ &\qquad \Big[+\hat{\mathbf{x}}_i(k|k) - \hat{\mathbf{x}}_{0j}(k|k)\Big]\\ &\qquad \cdot \hat{\mathbf{x}}_i(k|k) - \hat{\mathbf{x}}_{0j}(k|k)^T\Big\} \cdot \mu_{i|j}(k|k)\\ &filtering:\\ &\mu_j(k) = \frac{1}{c}\Lambda_j(k)\bar{c}_j(k)\\ &where,\ \Lambda_j(k) = N\left(r_j(k);\, 0,\, S_j(k)\right)\\ &combination:\\ &\hat{\mathbf{x}}(k|k) = \sum_j \hat{\mathbf{x}}_j(k|k)\mu_j(k)\end{aligned}$$

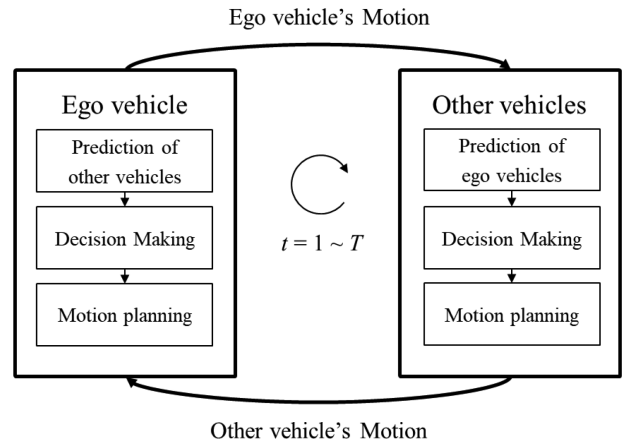

Fig. 3. Decision-making and motion planning process of vehicles.

$$\begin{aligned}\mathbf{P}(k|k) = \sum_j \Big\{&P_i(k|k) + [\hat{\mathbf{x}}_j(k|k) - \hat{\mathbf{x}}(k|k)]\\ &\cdot \Big[\hat{\mathbf{x}}_j(k|k) - \hat{\mathbf{x}}(k|k)\Big\} \cdot \mu_j(k|k)\end{aligned} \tag{8}$$

where c is the normalization factor for mode probability μ. Λ_j is the likelihood function of j-th model, which is a multivariate Gaussian density function. p_{ij} is a transition probability from i-th model to j-th one. r_j is the residual of the j-th model with measurement. S_j is the residual covariance. The details of the IMM are described in [43].

IV. Game Theory-Based Supervisor

Decision-making at intersections should take into account various traffic participants, such as vehicles, cyclists, and pedestrians. Even without considering Vulnerable Road User (VRU), more complex driving conditions should be taken into consideration. Fig. 3 illustrates the decision-making process of the vehicles that interact with each other. Drivers make their decisions based on given information such as the vehicle's position, velocity and the prediction of the target's future motion. In this step, the driver infers the intention of the other drivers to handle the uncertainty of the prediction. The desired motion is determined based on the predicted states of other vehicles and the current states of the ego vehicle. The motion planner of AV aims to find the desired trajectory and longitudinal behavior to avoid collisions and reach the destination. Therefore, the interactive nature of the decision-making and motion planning needs to be modeled to predict the future driving scene and decide the desired motion as accurately as possible. In other words, prediction and planning incorporate all traffic participants that can affect the behavior of the ego vehicle. Thus, prediction and motion planning cannot be done separately. This process is conducted up to the prediction horizon T to find the optimal maneuver of the ego vehicle, as shown in Fig. 3.

This study introduces the game theory to configure the supervisor module to make a decision based on the prediction results of the surrounding targets. Therefore, this approach assumed that each traffic participant decides their motion considering the

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probability of the maneuver and risk caused by the interaction with other participants. To formulate the game theory-based supervisor, definitions and notations are provided in Section IV-A. Then, a game and payoff function is formulated in Section IV-B and C.

A. Definitions and Notations

The types of behavior, which a vehicle can take at an intersection, are much more diverse than on main roads. Therefore, it is important to represent the behavior of intersection-approaching vehicles with a finite number of maneuvers. In the following, the driver's behavior is approximated by a finite set of maneuvers, $\mathcal{M}$. The i-th maneuver of the j-th vehicle at time step t, $m^t{}_{i,j}$, is defined as follows:

$$m_{i,j}^t \in \mathcal{M}_j^t \tag{9}$$

$\mathcal{M}_j^t$ means the maneuver set of the j-th vehicle. If the N_M maneuvers are defined for each vehicle, the maneuver set at the t-th time step is defined as follows.

$$\mathcal{M}_j^t = \left\{ m_{1,j}^t \quad m_{2,j}^t \quad \cdots \quad m_{N_M,j}^t \right\} \tag{10}$$

In this study, N_M is six, which combines lateral and longitudinal behaviors. Lateral behavior includes turning left, turning right, and going straight, while longitudinal behavior comprises crossing and stopping. If all vehicles share their intention and vehicle information, the optimal maneuver sets $\mathcal{M}_j^{t,*}$ can be determined by the control center or master vehicle of the intersection. The driving scene $\mathbb{S}^t$ at time step t is defined as follows:

$$\begin{aligned} \mathbb{S}^t &= \left\{ S_1^t \quad S_2^t \quad \cdots \quad S_{N_S}^t \right\} \\ &= \prod_j \mathcal{M}_j^t \end{aligned} \tag{11}$$

where $S^t{}_i$ means the i-th driving scene, which includes the maneuver sets of all vehicles. N_S is the number of the possible driving scene. Thus, N_S is N_V square of N_M. N_V is the number of a vehicle participating in the interaction.

However, directly obtaining the intention and information of other vehicles is difficult. In other words, the driving scene is not deterministic based on the sensor information alone. Therefore, a probabilistic approach is necessary to predict the future driving scene. Based on the probability of each maneuver of vehicles, the probability of the all-maneuver sets must be evaluated to assess the risk caused by the surrounding targets. Since the focus of this study is the maneuver decision of the ego vehicle, the uncertainty of the targets is taken into account to estimate the probability of the driving scene. The probability $\mathcal{P}_{E,i}^t$ of the $S^t{}_i$ and a set of the probability $\mathbb{P}_E^t$ are expressed as follows with the subscript E for the ego vehicle.

$$\begin{aligned} \mathcal{P}_{E,i}^t &= \prod_{\substack{n \\ n \neq E}} P\left(m_{i,n}^t\right) \\ \mathcal{P}_{E,i}^t \in \mathbb{P}_E^t &:= \left\{ \mathcal{P}_{E,1}^t \quad \mathcal{P}_{E,2}^t \quad \cdots \quad \mathcal{P}_{E,N_S}^t \right\} \end{aligned} \tag{12}$$

Based on the $\mathcal{P}_{E,i}^t$ and $\mathcal{M}_E^t$, the driving scene after the ego vehicle executes the specific maneuver is defined as follows:

$$S_E^t = \prod_{i=1}^{N_S} \mathcal{M}_E^t \times \mathcal{P}_{E,i}^t \tag{13}$$

$\mathcal{M}_E^t$ means the maneuver set of the ego vehicle. The planning problem is to evaluate the payoff U of the $S^t{}_E$ to find the optimal maneuver to manage the risk and improve traffic efficiency. The decision-making problem will be reformulated in the following two sections to solve the probabilistic game in uncontrolled intersections.

B. Decision-Making Strategy

The maneuvers of the vehicles are interdependent. The payoff function U should consider the possible risk from surrounding targets to avoid collisions. However, assessing only the risks to determine the driving mode can impede the flow of vehicles passing through the intersection. To address this issue, the time required for vehicles to pass through the intersection should also be taken into account in U. In this case, it is necessary to consider the maneuvers each vehicle will take and the traffic efficiency of the approaching vehicles at the intersection. For example, in a particular driving scene, even if the risk is deemed high, the probability of the driving scene occurrence may be low. Thus, to effectively determine the driving mode, the possibility of the driving scene occurrence should be considered along with risks and traffic efficiency. Moreover, the previous maneuver also impacts the decision of the current optimal maneuver. Thus, U of the $S^t{}_E$ can be evaluated using the maneuver evaluation function E for the maneuver of the ego vehicle and conditional probability for the driving scene as follows:

$$U(S_E^t) = E\left(m_{i,E}^t | S_i^t\right) \cdot P\left(S_i^t | \pi_E^{t-1}, x_E^t, x_T^t\right) \tag{14}$$

where, $x^t{}_E$ and $x^t{}_T$ are the states of the ego and target vehicles, respectively. $\pi^{t-1}{}_E$ is the sequence of the optimal maneuver of the ego vehicle from 1 to (t-1)-th time step. $m^t{}_{i,E}$, i-th maneuver of the ego vehicle at time step t.

Therefore, $U(S^t{}_E)$ can be obtained from the two probabilities. The first term of (14) represents the conditional probability of the $m^t{}_{i,E}$ given the driving scene $S^t{}_i$. Given the driving scene, the driver estimates the potential risk caused by the surrounding targets and tries to cross the intersection as quickly as possible. In other words, when the risk is lowered or the passing time is reduced, the probability of the $m^t{}_{i,E}$ is high. Therefore, the first term of (14) is modeled as a function of risk and traffic efficiency. The second term of (14) represents the conditional probability $S^t{}_i$ given the observation of the vehicles and maneuver of the ego vehicle in the previous time step. In this study, the IMM filter-based approach is introduced to infer the driving intention and predict the future motion of the other vehicles. Since the IMM filter estimates the probability of the possible maneuver of the vehicle given the observations, the outputs of the IMM filter can be used to evaluate the value of the second term.

The driving mode decision is illustrated in Fig. 4. $u_i{}^t$ is the value of payoff U when the ego vehicle executes the $m^t{}_{i,E}$ under S_i. The payoffs of all-possible cases are evaluated to determine

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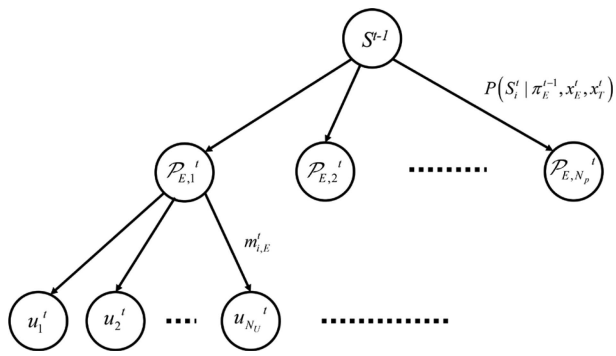

Fig. 4. Decision-making process.

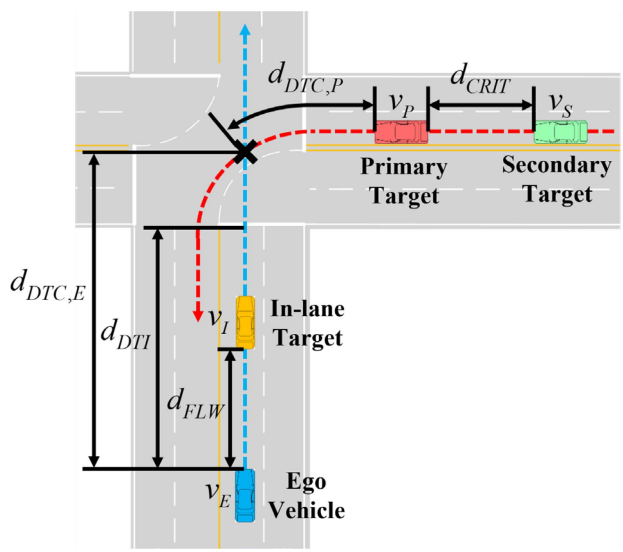

Fig. 5. Definition of the vehicle classification and variables to represent the driving scene.

the optimal driving mode. The details of the payoff function will be discussed in Section IV-C.

C. Payoff Function

U is composed of two functions, maneuver evaluation function E and probability for the driving scene P. E considers safety and traffic efficiency. Thus, E is composed of the risk term and traffic flow one as follows:

$$E = (1 - R) + F \tag{15}$$

To accurately evaluate risk R and traffic flow F, it is important to properly classify the approaching target vehicles in the intersection. In this study three categories of surrounding targets around the ego vehicle are identified, as illustrated in Fig. 5. The driving routes of the ego vehicle and target vehicles are represented as blue and red dashed lines. The first category is the in-lane target, which refers to the vehicle in front of the ego vehicle in the same lane. The second category is the primary target, which represents the vehicle at the front of the vehicle stream from the other branches of the intersection. The expected conflict point with the ego vehicle is used to determine the primary target. The conflict point is depicted as a black cross in Fig. 5. Finally, the third category is the secondary target, which refers to the following vehicle of the primary target. By properly classifying the surrounding targets, the driving scene in the intersection can be effectively analyzed, and the risk and traffic flow can be accurately evaluated.

After the primary target passes through the expected conflict point, the secondary target is redefined as the primary target. Then, the following vehicle of the previous secondary target is defined as the secondary target. This is because the intersection crossing problem at uncontrolled intersections is summarized as whether to cross first, pass a primary target and cut off the vehicle stream, or pass all targets and decide again with new targets. This is in line with the analysis of gap acceptance behavior in the unsignalized intersection theory. This classification can effectively represent the vehicle stream and define the safety parameters shown in Fig. 5.

First, d_{DTI} and d_{DTC} mean the remaining distance to the stop line of the intersection and the conflict point, respectively. d_{FLW} means the clearance between the in-lane target and the ego vehicle. To apply the uncontrolled intersection theory to the payoff function, the distance between the primary and secondary target is defined as d_{CRIT}. *TTC* and clearance *C* are widely used as indices for risk assessment in the design of ADAS functions. Since the conventional *TTC* and *C* are defined with respect to the preceding vehicle, it is difficult to assess the risk due to the intersection approaching targets. Thus, *TTC* and *C* are modified to take into account the primary target and the conflict point as follows [44]:

$$C_{conf} = d_{DTC,E} + d_{DTC,P} \tag{16}$$

$$TTC_{conf} = \frac{d_{DTC,E}}{v_E} + \frac{d_{DTC,P}}{v_P} \tag{17}$$

where $d_{DTC,E}$ and $d_{DTC,P}$ are the d_{DTC} of ego vehicle and primary target. v_E and v_P mean the velocity of the ego vehicle and primary target. C_{conf} and TTC_{conf} reflect that collision avoidance at conflict points can be achieved by increasing the *TTC* of any one vehicle.

The critical gap t_{CRIT} and follow-up gap t_{FLW} are defined as follows:

$$t_{CRIT} = \frac{d_{CRIT}}{v_S} \tag{18}$$

$$t_{FLW} = \frac{d_{FLW}}{v_I} \tag{19}$$

where v_S and v_I are the velocity of the secondary and in-lane targets. To represent traffic efficiency, the remaining time to enter the intersection is defined as follows:

$$t_{DTI} = \frac{d_{DTI}}{v_E} \tag{20}$$

Since the behavior of the vehicle has uncertainty, the payoff function should consider the future motion of each vehicle. The local model of the IMM filter for each maneuver is used to predict the future motion of the vehicles for 2 s with a sampling time of

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TABLE I
PARAMETERS FOR PAYOFF FUNCTION

Parameter	Value	Index	Value
k_{TTC}	3	$TTC_{th}\,(s)$	5
k_C	3	$C_{th}\,(m)$	15
k_{CRIT}	4	$t_{CRIT,th}\,(s)$	2
k_{FLW}	8	$t_{FLW,th}\,(s)$	4
k_{DTI}	2	$t_{DTI,th}\,(s)$	5

0.1 s. The R and F at k-th sampling time are defined as follows:

$$R(k) = -k_{TTC}\ln\left(\frac{TTC_{conf}(k)}{TTC_{th}}\right) - k_c\ln\left(\frac{C_{conf}(k)}{C_{th}}\right) - k_{CRIT}\ln\left(\frac{t_{CRIC}(k)}{t_{CRIC,th}}\right) + \left(\exp\left(\frac{t_{FLW}(k)-t_{FLW,th}}{k_{FLW}}\right)-1\right) \tag{21}$$

$$F(k) = \exp\left(\frac{(t_{DTI}(k)-t_{DTI,th})}{k_{DTI}}\right)-1 \tag{22}$$

where, k_{TTC}, k_c, k_{CRIT}, k_{FLW}, and k_{DTI} are the weights to adjust the effect of each term of R and F. The risk assessment uses a logarithmic function to evaluate TTC_{conf}, C_{conf}, and t_{CRIT}, where smaller values indicate higher risk. For t_{FLW} and t_{DTI}, an exponential function was used to assess the risk and traffic efficiency, where smaller values indicate lower risk and better traffic efficiency. However, when an index exceeds a certain threshold, further assessment is unnecessary. Safety indexes with thresholds are used in this study to limit the range of risk and traffic efficiency evaluation for exponential and logarithmic functions. The weight of each term of R and F is adjusted by the parameter k_{TTC}, k_c, k_{CRIT}, k_{FLW}, and k_{DTI}.

Finally, the E for a specific driving scene is calculated by a weighted sum of the $R(k)$ and $F(k)$ with the exponential function as follows:

$$E = \sum_{k=1}^{N_p}\exp\left(-k_d\cdot k\cdot dt\right)\left(1-R(k)+F(k)\right) \tag{23}$$

where, N_p and dt are the prediction horizon and the sampling time, respectively. k_d is the decay rate for the prediction horizon. The dt is 0.1 s, N_p is 20. The parameters for the payoff function are summarized in Table I. The value of the thresholds is derived from the analysis of the driving data [44], [45], [46].

The probability of the driving scene is calculated from the maneuver probability of the IMM filter as follows:

$$P\left(S_i^t|\pi_E^{t-1}, x_E^t, x_T^t\right) = \prod_{\substack{n\\ n\neq E}}\mu_{j,n} \tag{24}$$

where $\mu_{j,n}$ is the probability of the j-th maneuver of the n-th vehicle. The maneuver set of each target vehicle is defined by the intersection structure.

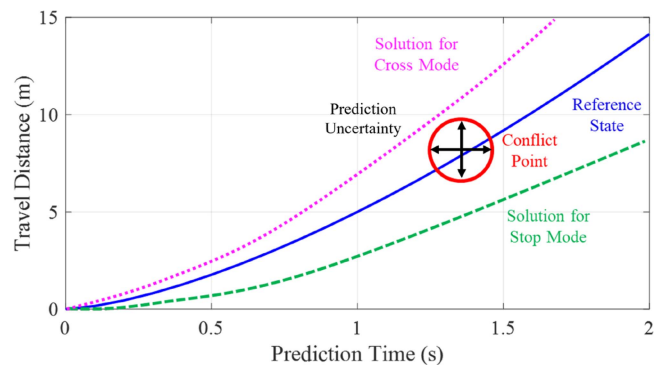

Fig. 6. The concept of the reference state decision within the prediction horizon.

V. STOCHASTIC MPC-BASED MOTION PLANNING

The objective of the motion planner is to determine the optimal control input based on various information. The longitudinal acceleration is calculated by using the SMPC approaches based on the intention of the surrounding targets and the driving mode. The motion planner is composed of two modules: reference state decision, and SMPC. The reference state decision module generates reference states within the prediction horizon of the SMPC. Then, the SMPC problem is formulated to track the reference state while securing the safety of the ego vehicle and satisfying the constraints.

A. Reference State Decision

Before optimizing the desired motion, the reference states are determined based on the predicted conflict point. The concept of the reference state decision is described in Fig. 6 [44]. The reference state is determined within the prediction horizon to define the SMPC for reference tracking. First, the travel distance is predicted assuming that there is no target vehicle. To prevent unavoidable collisions when approaching an intersection at excessive speed, the approach mode was applied to the reference state decision [44]. The approach mode reduces the velocity below the speed limit to prepare for the possibility of a target vehicle being in a blind spot. The reference without a target is marked as a blue solid line in Fig. 6. Then, the conflict point exists on the reference without a target, which is described as a red circle of arbitrary size. The size of the conflict point is determined by the prediction uncertainty of the targets, and as mentioned in Section III, the IMM is defined by using (1) to (3). The covariance of the IMM is used to determine the size of the conflict point as a probabilistic chance constraint of SMPC. The conceptual solution for cross and stop mode is represented as a magenta dotted line and green dashed line in Fig. 6. Since the safety parameters, such as TTC and C, are correlated with the states of both vehicles, it is difficult to directly derive the solution. Thus, correlated optimization for ego and target vehicle states is required.

The objective of the proposed motion planner is to avoid collision with other targets while passing the intersection as quickly as possible, which applies equally to both cross and stop

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modes. However, passing the intersection excessively fast can increase the risk of colliding with other vehicles coming from the blind spot. Therefore, it is appropriate to minimize the change in the states from the non-target case. The reference without a target is used as a reference state for SMPC under cross mode. The same model, which is described in Section III-A, is used to generate the reference state. For longitudinal motion, v_{des} is set as the speed limit of the road, and the in-lane target is also considered to determine the acceleration inputs.

B. SMPC Design

A motion planner needs to consider various uncertainties to manage the risk caused by the surrounding objects. For example, sensor noise, perception accuracy, and actuator performance are representative uncertainties of motion planning. Therefore, it is necessary to use a robust motion planner to consider these uncertainties. In this study, SMPC is utilized to consider the perception and prediction uncertainties in determining longitudinal acceleration. A chance constraint is introduced in SMPC to update the uncertainty at every moment. The prediction uncertainty is acquired from the covariance of the perception module, which is based on the extended Kalman filter. The process update of the IMM filter is used to evaluate the probability of the possible maneuver of the target vehicles and estimate the prediction uncertainty.

To avoid collision in intersections, using longitudinal motion is much safer than lateral motion. This is because a rapid change in the driving route within the intersection could cause a collision with other vehicles. Therefore, the SMPC is designed under the assumption that the ego vehicle will follow the predefined driving route. The longitudinal model with actuator delay is used to define the plant model of SMPC. The actuator delay is modeled as a first-order delay model with the following state vector $x(k)$ and input vector $u(k)$:

$$\begin{aligned} x(k) &= \begin{bmatrix} p(k) & v(k) & a(k) \end{bmatrix}^T \\ u(k) &= a_{input}(k) \end{aligned} \tag{25}$$

where $p(k)$, $v(k)$, and $a(k)$ are the driving distance, velocity, and acceleration on the curvilinear coordinate along the reference path of the ego vehicle. $a_{input}(k)$ is the longitudinal acceleration input, which is generated by the accelerating and braking of the ego vehicle. The longitudinal model for the SMPC is defined as follows:

$$x(k+1) = \begin{bmatrix} 1 & dt & 0 \\ 0 & 1 & dt \\ 0 & 0 & 1 - dt/\tau_x \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 0 \\ dt/\tau_x \end{bmatrix} u(k) \tag{26}$$

dt is the sampling time. τ_x is the constant of the actuator time delay.

The purpose of the proposed motion planner is to track the reference states. Thus, the objective function of the SMPC is defined as follows:

$$J = \sum_{k=1}^{N_p} \mathrm{E}\left(\|x(k) - x_{ref}(k)\|_Q^2 + \|u(k)\|_R^2 \right) \tag{27}$$

The first term of J replaces the state regulation term to minimize the error of the reference tracking. Therefore, J has the advantage of not having to define error dynamics. Q is the weight matrix for the reference tracking. The second term is used to minimize the control effort. R is the weight matrix for the input minimization. N_p is the number of the prediction steps of the SMPC problem.

The initial condition of the SMPC is defined as the current state of the ego vehicle.

$$x(1) = \begin{bmatrix} 0 & v_x & a_x \end{bmatrix}^T \tag{28}$$

where v_x is the longitudinal velocity and a_x is the acceleration of the ego vehicle when solving the SMPC problem.

The SMPC takes into account the uncertainty of the states by introducing a chance constraint. As the MPC framework considers future states for the tracking problem, the prediction results of surrounding targets are included in the constraint definitions. Therefore, constraints related to the behavior of the target vehicle are defined based on the probability of the predicted states. In this study, the uncertainty of the target prediction is used to define the chance constraints for the ego vehicle's position. First, the position constraints are defined as inequalities based on the driving mode.

$$\begin{aligned} Cross\ Mode:\ & p(k) - C(k) \geq p_{tar}(k) \\ Stop\ Mode:\ & p(k) + C(k) < p_{tar}(k) \end{aligned} \tag{29}$$

where $p_{tar}(k)$ is the position prediction result of the target vehicle by using the IMM filter. The safety margin $C(k)$ is calculated by using the predicted states of the target vehicle. Since the ego vehicle under cross mode should pass the conflict point earlier than the targets, the lower boundary is only defined for the position constraint. On the contrary, the upper boundary is applied to make the ego vehicle cross the conflict point later. $C(k)$ is defined as follows:

$$C(k) = \left(TTC_{\min} - \frac{p_{tar}(k)}{v_{tar}(k)} \right) v(k) + C_{\min} \tag{30}$$

where $v_{tar}(k)$ is the predicted velocity of the target vehicle by using the IMM filter. $C(k)$ is designed to maintain $TTC_{\min}$ and $C_{\min}$ with the target vehicle. If multiple targets exist, position constraint is applied for all targets to prepare for possible future risk. $p_{tar}(k)$ and $v_{tar}(k)$ have a probabilistic distribution, which increases as the IMM filter predicts the far future. Therefore, the position constraint is redefined as the chance constraint as follows:

$$\begin{aligned} Cross\ Mode:\ & \mathrm{P}\left(p(k) - C(k) \geq p_{tar}(k)\right) > \beta \\ Stop\ Mode:\ & \mathrm{P}\left(p(k) + C(k) < p_{tar}(k)\right) > \beta \end{aligned} \tag{31}$$

where β is the threshold for the chance constraint. The probability distribution is calculated for each k-th prediction step. The covariance of the IMM filter is used to define the probability of (32) with multivariate Gaussian distribution. Since the state of the IMM filter consists of the x, y, θ, and v, not only the prediction error for velocity but also the position error with respect to the intersection map are considered in the (32). Thus, the chance

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TABLE II
PARAMETERS FOR SMPC

Parameter	Value	Parameter	Value
$dt\ (s)$	0.1	$\tau_x\ (s)$	0.5
N_P	20	$TTC_{\min}\ (s)$	2
$C_{\min}\ (m)$	5	β	0.9
$v_{x,\max}\ (m/s)$	20	$a_{y,max}\ (m/s^2)$	3
$a_{x,\min}\ (m/s^2)$	-3	$a_{x,\max}\ (m/s^2)$	1
$\dot{a}_{x,\max}\ (m/s^3)$	10		

TABLE III
INITIAL CONDITION FOR CASE STUDY

Vehicle	Case #1			Case #2		
	$d_{DTI,ini}$ (m)	v_{ini} (m/s)	$v_{\max}$ (m/s)	$d_{DTI,ini}$ (m)	v_{ini} (m/s)	$v_{\max}$ (m/s)
Ego	100	7	12	75	10	12
Primary #1	100	8	12	50	11	12
Primary #2	-	-	-	125	11	12
Secondary	150	8	12	100	11	12
Tertiary	-	-	-	150	10	12
In-lane	50	10	12	-	-	-

constraint is adjusted based on the performance of the perception and prediction modules.

The deterministic constraints are defined for the state of the ego vehicle as follows:

$$0 < v_x(k) < \min\left(\left[\begin{array}{cc} v_{x,\max} & \frac{a_{y,\max}}{\gamma(k)} \end{array}\right]\right) \tag{32}$$

$$a_{x,\min} < a_x(k) < a_{x,\max} \tag{33}$$

$$\frac{|a_x(k+1) - a_x(k)|}{dt} < \dot{a}_{x,\max} \tag{34}$$

Equations (33), (34), and (35) are the constraints for longitudinal velocity, acceleration, and jerk, respectively. $v_{x,\max}$, is the maximum longitudinal velocity. Assuming that the lateral velocity is ignorable, the safe longitudinal velocity is calculated from the maximum lateral acceleration $a_{y,\max,}$ and yaw rate $\gamma(k)$. $a_{x,\min,}$ and $a_{x,\max}$ are the minimum and maximum longitudinal acceleration, respectively. $\dot{a}_{x,\max}$ is the maximum longitudinal jerk. The parameters for SMPC are summarized in Table II. The proposed SMPC problem is solved by FORCES in the MATLAB/Simulink environment [47].

VI. RESULTS

A simulation with multiple target vehicles was carried out to evaluate the performance of the proposed algorithm. To compare the performance, a base algorithm is designed and used to obtain the reference results. The base algorithm uses the same motion planner as the proposed algorithm. However, the supervisor is replaced by a rule-based algorithm, which is designed based on the principle of first-in-first-out rules. The target vehicles in the simulation study are governed by a base algorithm. When approaching the intersection, the approach mode was applied to both algorithms. The approach mode is deactivated when the ego vehicle is in the cross mode without blind spots or when the driving mode is changed to stop mode. To show the effectiveness of the proposed approach, two case studies and a Monte-Carlo simulation were conducted.

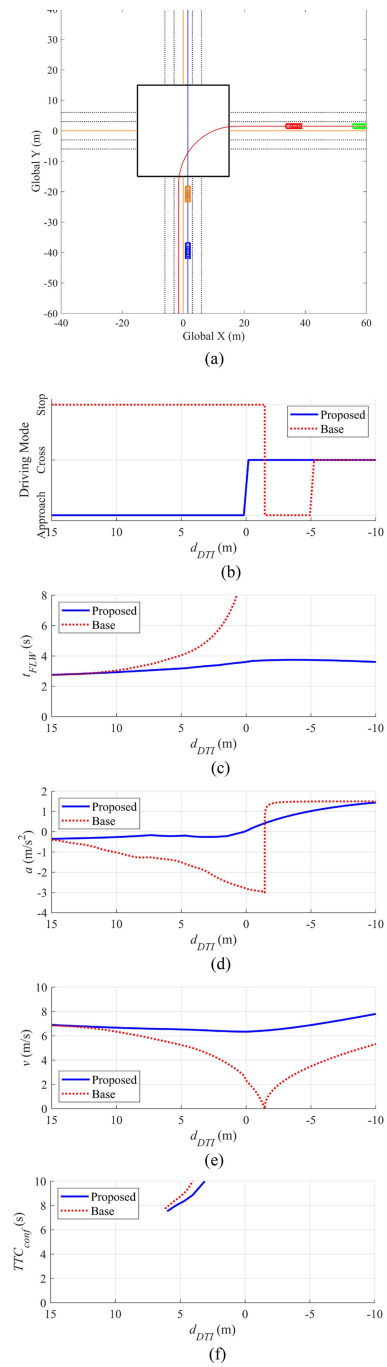

Fig. 7. Simulation results of Case #1. (a) Simulation scenario. (b) Driving mode history. (c) Follow-up gap history. (d) Acceleration history. (e) Velocity history. (f) TTC history.

A. Case Study

In the intersection, a variety of scenarios can occur due to the complex interaction of the vehicles. Among these scenarios, two cases are selected to compare the proposed and base algorithms. The first case, denoted as Case #1, is the scenario in which the in-lane, primary, and secondary targets approach the intersection. The approach timing is set so that the in-lane target approaches

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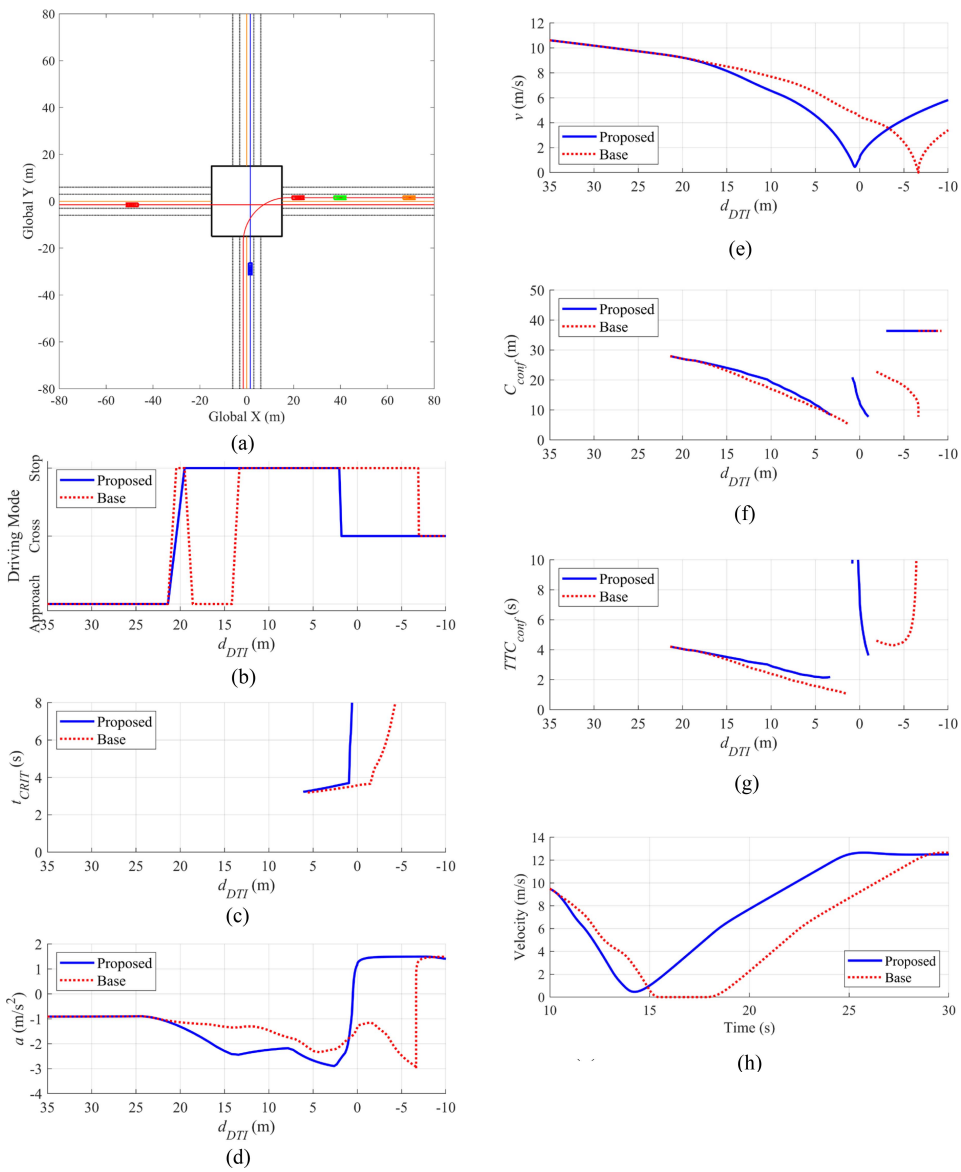

Fig. 8. Simulation results of Case #2. (a) Simulation scenario. (b) Driving mode history. (c) Critical gap history. (d) Acceleration history. (e) Velocity history. (f) Clearance history. (g) TTC history. (h) Velocity history in the time domain.

the intersection earlier than the primary target. Thus, the ego vehicle is required to determine whether to pass the intersection following the in-lane target.

Case #2 is a scenario with multiple targets designed to evaluate the proposed algorithm under a complex driving scenario. Primary targets #1 and #2 approach the intersection from different branches, while the secondary and tertiary targets behind primary target #1 constitute the vehicle stream. The initial condition of the ego vehicle is set to enter the intersection at the intermediate timing between primary target #1 and the secondary target. Then, primary target #2 and tertiary target approach the intersection, requiring the geo vehicle to determine how to respond to the targets from a different branch and the vehicle stream simultaneously. By conducting case studies on cases #1 and #2, the proposed algorithm's performance can be evaluated for complex scenarios at uncontrolled intersections. The initial condition of each case is summarized in Table III.

The simulation results of the Case #1 are depicted in Fig. 7. Fig. 7(a) shows the scene of Case #1, with the primary, secondary, and in-lane targets depicted as orange, red, and green vehicles, respectively. The ego vehicle is marked as a blue vehicle. As mentioned before, the in-lane target is set to arrive at the intersection earlier than the primary target. The results of the driving mode decision in this scenario are shown in Fig. 7(b). The primary target is detected at $d_{DTI} = 15$ m. Before detecting the target, the ego vehicle is the approach mode. Thus, mild braking of about -0.3 m/s^2 is applied to prevent the unavoidable collision as shown in Fig. 7(d). Since the supervisor of the base algorithm is based on the first-in-first-out rule considering the t_{DTI}, the driving mode of the base algorithm is changed to stop mode. Although t_{FLW} is about 2.4 s, the base algorithm decides not to follow the in-lane target. The ego vehicle decelerates and approaches the intersection slower than the primary as well as secondary targets. As a result, the ego vehicle stopped at $d_{DTI} = -2$ m and passed all targets.

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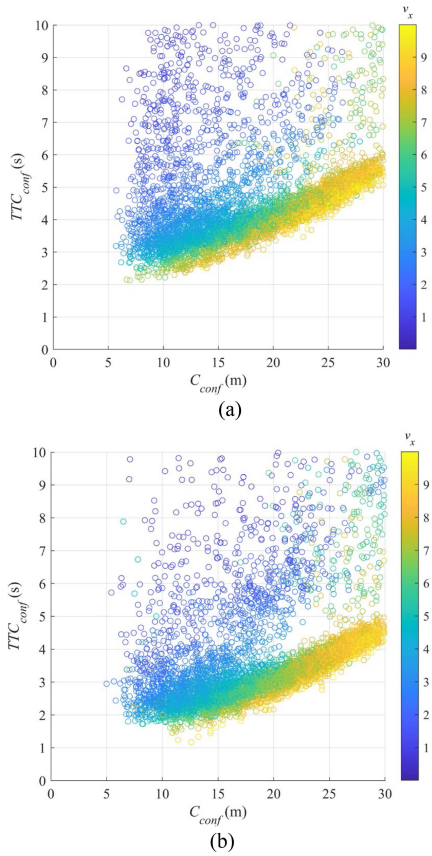

Fig. 9. TTC_{conf} - C_{conf} distribution. (a) Proposed algorithm. (b) Base algorithm.

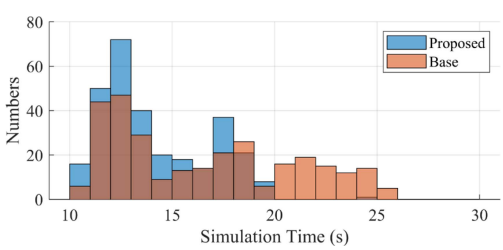

Fig. 10. Simulation time when the ego vehicle passes the intersection.

Meanwhile, the ego vehicle controlled by the proposed algorithm maintains the approach mode to follow the in-lane target. The supervisor interprets the intention of the primary target, which yields the in-lane target. Thus, the supervisor predicts that there is a high possibility of yielding to the ego vehicle that follows the in-lane target at a close distance. The t_{FLW} is maintained at a reasonable level as shown in Fig. 7(c). After the ego vehicle enters the intersection and secures the FOV, the driving mode is changed to cross mode. As shown in Fig. 7(f), TTC_{conf} is maintained above the safety level for both cases. Therefore, the decision-making module of the proposed algorithm achieves safety and traffic efficiency, simultaneously.

The simulation results of Case #2 are described in Fig. 8. The scenario is depicted in Fig. 8(a) and the history of the driving mode is displayed in Fig. 8(b). In the simulation results of the base algorithm, the ego vehicle detected the primary target #1 at $d_{DTI} = 22$ m. As primary target #1 approaches the intersection earlier than the ego vehicle, the ego vehicle determines the stop mode. However, the base supervisor misinterpreted the behavior of primary target #1 as a yield to the ego vehicle. Consequently, the driving mode is changed from stop mode to approach mode to cross the intersection earlier than the primary target #1. Due to this misjudgment, TTC_{conf} decreased faster than that of the proposed algorithm as shown in Fig. 8(g). Thus, the driving mode is changed again to stop at $d_{DTI} = 13$ m to manage the risk. As shown in Fig. 8(d) and (e), since the deceleration timing is late, v_x is not reduced to a safe level. Thus, the TTC_{conf} and C_{conf} eventually reduced to 1.15 s and 5.43 m, which is a risky level. As shown in Fig. 8(c), t_{CRIT} between primary target #1 and secondary target increases to 4 s, similar to that of the proposed algorithm. However, t_{CRIT} between the secondary target and tertiary target, depicted after $d_{DTI} = -2$ m, slowly increased. This is because the hard braking of the ego vehicle was inferred as a yield sign to the tertiary target. Thus, t_{CRIT} did not increase sufficiently, and the ego vehicle could not cut the vehicle stream to avoid a collision. Consequently, the ego vehicle completely stopped after crossing the stop line, taking a long time to cross the intersection, as shown in Fig. 8(h).

Meanwhile, the proposed algorithm exhibited a different behavior than the base algorithm. Initially, the driving mode of the ego vehicle was changed to cross as primary target #1 decelerated to prepare for a left turn. However, the intention of primary target #1 was inferred immediately, and the driving mode was changed to stop. After the secondary target was detected at $d_{DTI} = 6$ m, t_{CRIT} was continuously evaluated and increased. In this case, the proposed algorithm changed the driving mode from stop to cross at $d_{DTI} = 2$ m. After primary target #1 passed the conflict point, t_{CRIT} was evaluated by using secondary and tertiary targets. Since the ego vehicle already started reaccelerating, the tertiary target yielded the ego vehicle. This is evident from the rapid rise of the t_{CRIT}. Primary target #2 did not cause a deadlock by waiting for other vehicles after stopping. As shown in Fig. 8(h), the ego vehicle governed by the proposed algorithm crossed the intersection earlier than the base algorithm. TTC_{conf} and C_{conf} of each algorithm are depicted in Fig. 8(f) and (g). In the case of the base algorithm, the decrease of TTC_{conf} and C_{conf} was not managed and continues to reduce to a risky level. On the other hand, TTC_{conf} and C_{conf} are well managed by the proposed algorithm. Therefore, it can be concluded that the proposed algorithm prevents the occurrence of dangerous situations while reducing the time required to pass the intersection.

B. Monte-Carlo Simulation

To validate the proposed approach in various conditions, the Monte-Carlo simulation is conducted. The initial conditions of the Monte-Carlo simulation are generated by using the following normal distributions.

$$\begin{aligned} d_{DTI,\text{ini}} &= N(100\,\text{m},\ 20\,\text{m}) \\ v_{\text{ini}} &= N(10\,\text{m/s},\ 5\,\text{m/s}) \\ v_{\max} &= N(12\,\text{m/s},\ 3\,\text{m/s}) \end{aligned} \tag{35}$$

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For the ego vehicle and in-lane target, the input nodes are set to the same. Similarly, the primary and secondary targets have the same input nodes. The exit node of the vehicles is randomly selected except for the same branch of the input node. To maintain the intersection approach order, the following conditions are additionally applied.

$$\begin{aligned} d_{DTI,ini,I} &= \min\left(d_{DTI,ini,I},\ d_{DTI,ini,E} - 25\text{m}\right) \\ d_{DTI,ini,P} &= \min\left(d_{DTI,ini,P},\ d_{DTI,ini,S} - 25\text{m}\right) \end{aligned} \tag{36}$$

Furthermore, the actuation delay τ_x is generated for each simulation to provide a different driving behavior of the targets as follows:

$$\tau_x = N(0.5\,\text{s},\ 0.03\,\text{s}) \tag{37}$$

The simulation results are summarized in Figs. 9 and 10. The distribution of the safety parameters, TTC_{conf} and C_{conf}, is displayed with a color marker based on v_x. As the vehicles approach each other, the TTC_{conf} and C_{conf} decrease toward the origin of the TTC_{conf} vs. C_{conf} plane. Braking is applied to manage the risk and increase the TTC_{conf}. Thus, the trajectory of each simulation is shown as a 'C' shape on TTC_{conf} vs. C_{conf} plane.

As shown in Fig. 9(a), the TTC_{conf} and C_{conf} distribution of the proposed algorithm is maintained above the constraint of the SMPC. In addition, since the trajectory is concentrated in a 'C' shape, unnecessary deceleration to increase the TTC_{conf} hardly occurs. On the contrary, the base algorithm violates the constraint of SMPC as shown in Fig. 9(b). In particular, TTC_{conf} is reduced up to 1.1 s, which is a highly dangerous level. In some cases, unnecessary braking is applied and TTC_{conf} is maintained high even when C_{conf} is sufficient. For this reason, the spacing angle of the 'C' shape is reduced. Therefore, the proposed algorithm guarantees performance under various driving conditions without unnecessary control input.

The proposed algorithm's efficiency was compared to the base algorithm by analyzing the simulation time it takes for the ego vehicle to pass through the intersection. The results are summarized in Fig. 10 as a histogram. Since the intersection size and the initial position of the ego vehicle affect the simulation time, the bin edges start at 10 s. The histogram shows that most cases of the proposed algorithm passed the intersection within 20 s, with very few cases taking about 25 s. The shape of the distribution between the base and the proposed algorithm is similar within the interval of 20 s, but the distribution of the base algorithm is smaller than the proposed one within 15 s. Furthermore, about 25% of the results from the base algorithm are located in a region of more than 20 s, and almost evenly distributed over 15 s. Therefore, it can be concluded that the proposed algorithm improves traffic efficiency while ensuring safety.

VII. Conclusion

A motion planner was proposed for uncontrolled intersections using a game theory-based approach. Based on the sensor measurements, the probability of the possible maneuver of the targets was predicted by using an interacting multiple-model filter. The process update of the filter is based on the intelligent driver model. The probability of the driving scene was estimated based on the prediction results of each target. The supervisor of the motion planner was designed based on game theory. To evaluate the payoff, the probability of the driving scenes and uncontrolled intersection theory were integrated to define decision-making as a probabilistic game. The safety parameters, such as time to collision, clearance, follow-up gap, and critical gap were defined to evaluate the risk caused by surrounding targets. The stochastic model predictive controller was utilized to consider the probabilistic behavior of the targets when determining the optimal longitudinal acceleration. The simulation study was conducted to evaluate the proposed algorithm. The simulation results showed that the proposed algorithm achieves risk management and reduction of travel time, simultaneously.

Future works of the motion planner for the uncontrolled intersection can focus on three aspects. Firstly, improving the prediction using learning-based approaches can be considered. A learning-based predictor can be applied not only to predict the behavior of individual targets but also to interpret the driving scene. Secondly, the payoff function can be extended to consider more complex driving scenarios. Additionally, the introduction of connected vehicles and infrastructure can provide additional information to enhance risk assessment. Lastly, the personalization of the motion planner can be explored. Since drivers have different driving characteristics and acceptable levels of risk, adapting the driving characteristics of the planner is important to enhance driver acceptance of autonomous driving.

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